

Jainta Paul

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Summary

Former Software Engineer - Pursuing Ph.D. in Computer Science. Built static-analysis platforms, automated security pipelines, and program-analysis tools, and led end-to-end development of systems involving OpenPLC/LLVM instrumentation, machine-learning evaluation pipelines, and agent-based cyber-physical experimentation. Proficient in Python, Java, C++, and JavaScript, with strong experience in software development, automation, security, and systems engineering.

Education

University of Utah, Kahlert School of Computing

Ph.D. Student in Computer Science, Ongoing

Jan 2024 – Present

Salt Lake City, UT

University of Utah, Kahlert School of Computing

M.S. in Computer Science, GPA: 4.00/4.00

May 2026

Salt Lake City, UT

Bangladesh University of Engineering and Technology

B.Sc. in Computer Science and Engineering, GPA: 3.32/4.00

Feb 2017 – May 2022

Dhaka, Bangladesh

Technical Skills

Languages: Python, Java, C++, JavaScript, SQL

Systems & Tools: Git, Docker, Linux, CUDA, PyTorch

Software/Security: Static Analysis, Taint Analysis, Vulnerability Detection, Threat Modeling

Software Engineering Experience

Software Engineer

Jun 2022 – Nov 2023

[OpenRefactory, Inc.](#)

- **Test Drive – OpenRefactory:** Designed and led a GitHub-integrated web platform that allowed users to run static-analysis trials on real-world repositories, with OAuth authentication, repository selection, automated job orchestration, and result reporting.
- **Open-Source Bug Discovery Pipeline:** Built automated workflows to analyze open-source repositories, identify security and compliance issues, triage findings, and generate reports for maintainers, reducing the amount of manual review needed in the vulnerability reporting process.
- **Intelligent Code Repair:** Improved the Java Intelligent Code Repair engine by implementing call-graph construction, points-to analysis, and static taint tracking, expanding its ability to detect vulnerabilities and generate more precise repairs across large codebases.

Selected Research Projects

ICSTRACKER: Intrusion Backtracking for Industrial Control Systems

DSN 2025

Paper: [IEEE Xplore](#)

Code: [GitHub](#)

- Developed a provenance-based intrusion backtracking framework for Industrial Control Systems (ICS) that reconstructs attack paths from physical process anomalies to their originating cyber events.
- Built and instrumented the MiniWater OpenPLC testbed, integrating C/C++, LLVM, and Python-based tracing to capture controller execution and runtime events.
- Designed data collection and analysis pipelines for generating structured execution traces to support cyber-physical attack attribution.

Sensor Privacy as a Spectrum: Quantifying Privacy in Edge and Multimodal Systems through Games *PETS 2026*

Paper: [PoPETS](#)

Code: [GitHub](#)

- Introduced a game-based adversarial privacy framework demonstrating that highly constrained on-device outputs, including binary event streams, can act as privacy-leakage channels from which adversaries infer sensitive user activities.
- Built end-to-end adversarial inference pipelines that exploit temporal coherence, sensor placement, accessible modalities, and auxiliary context, achieving approximately 65% activity-inference accuracy from highly compressed outputs—about 4× random guessing.
- Implemented and evaluated multiple adversary models, including Decision Trees, Random Forests, CNNs, LSTMs, and Transformers, across four motion-sensing datasets to quantify how privacy leakage changes with output granularity,

sampling rate, channel availability, and temporal context.

- Evaluated cross-modal representation attacks showing how shared representations can make low-sensitivity IMU signals predictive of private semantic attributes associated with hidden high-fidelity modalities.

ICS-AGENT: Agent-Based Wargaming for Industrial Control Systems Cybersecurity

2025 – Present

- Introduced a bounded cyber-physical wargaming framework for controlled cybersecurity experimentation in industrial control systems, modeling attacker–defender interaction as a sequence of observations, decisions, actions, and physical-process consequences.
- Designed attacker and defender interfaces that independently control system knowledge, runtime observability, memory, and action authority, enabling repeatable experiments that isolate how attacker knowledge and defender capability affect cyber-physical safety.
- Built and evaluated rule-based and agentic attacker–defender workflows across realistic water-treatment and water-distribution ICS testbeds, spanning 29 attack scenarios and hundreds of seeded experimental runs while measuring detection, intervention, containment timing, safety violations, and operational disruption.
- Demonstrated a detection-to-protection gap in ICS defense: attacks can still drive physical processes into unsafe states despite successful detection and defender intervention, revealing when system knowledge, process timing, and operational leverage dominate defensive response.

Towards Cross-Physical-Domain Threat Inference for Industrial Control System Defense Adaptation

ACM

CCS-RICSS 2024

Paper: [ACM Digital Library](#)

- Developed a threat inference framework that correlates cyber attacks with physical process behavior to improve adaptive defense in industrial control systems.
- Modeled cross-domain attack propagation using process semantics to support earlier detection and mitigation of cyber-physical threats.

HyTwin: A Digital Twin Framework for Hybrid Systems Verification

NASA Formal Methods (NFM) 2025

Paper: [Springer Nature](#)

- Contributed to a digital twin framework for hybrid systems that combines runtime system observations with formal verification models.
- Assisted in implementing verification workflows for reasoning about cyber-physical system behavior under dynamic operating conditions.

Service & Awards

- Research Mentor, University of Utah REU, Summer 2024.
- Artifact Evaluator, USENIX VehicleSec 2026 and ACSAC 2025.
- Graduate Tuition & Research Scholarship, University of Utah; University Admission Scholarship, BUET.

References

Dr. Luis Garcia

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Dr. Pratik Soni

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